2ND SEMESTER LAB REPORT

TITLE : IDENTIFICATION OF RESISTOR COLOR CODES



Faiq Ali (249520)

Muhammad Zeeshan (243346)

Muhammad Waqas (240756)

BACHELOR OF COMPUTER ENGINEERING (Fall-2025)

Submitted To:

Engr. Shanzay amil

Lab Engineer

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
FACULTY OF ENGINEERING AIR UNIVERSITY,ISLAMABAD

	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 2

Lab Title: IDENTIFICATION OF RESISTOR COLOR CODES

Objective: 1. To find the value of a resistor and its tolerance by color-coding.

Student Name: Muhammad Waqas -Muhammad Zeesha -Faiq Ali

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 22/feb/2025

Signature: _____

LABORATORY EXPERIMENT NO. 2

IDENTIFICATION OF RESISTOR COLOR CODES

OBJECTIVE:

1. To find the value of a resistor and its tolerance by color-coding.
2. To measure the value of the resistor by Digital Multi-Meter (DMM).

Four Band Resistor



The four band resistor indicates the first digit for the 1st band, second digit for the 2nd band, third digit for 3rd band (multiplier or power of 10) and the fourth digit for 4th band represent tolerance of ±5%, ±10%, and ±20%.

Band	Purpose
1	First significant digit
2	Second significant digit
3	Multiplier of power of 10
4	Tolerance

Example:

1st Band – Red (2)

2nd Band – Orange (3)

3rd Band – Green (10^5)

4th Band – Gold (±5%)

<<<<<<<<<<<<<<<<<<<<<(Lab Tasks)>>>>>>>>>>>>>>>>>>>>>

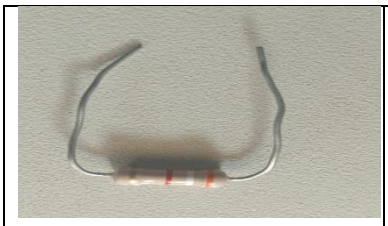
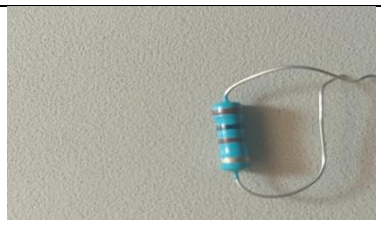
1. **Given the nominal values and tolerances in the given Table, determine and record the corresponding color code bands.**

Value	Band 1	Band 2	Band 3	Band 4
27 @ 10%	Red	Purple	Black	Silver
56 @ 10%	Green	Blue	Brown	Silver
180 @ 5%	Brown	Grey	Brown	Gold
390 @ 10%	Orange	White	Brown	Silver
680 @ 5%	Blue	Grey	Brown	Gold
1.5 k @ 20%	Brown	Green	Red	No band
3.6 k @ 10%	Orange	Blue	Red	Silver
7.5 k @ 5%	Purple	Green	Red	Gold
10 k @ 5%	Brown	Black	Orange	Gold
47 k @ 10%	Yellow	Purple	Orange	Silver
820 k @ 10%	Grey	Red	Black	Silver
2.2 M @ 20 %	Red	Red	Green	No band

2. **Given the color codes in the given Table, determine and record the nominal value, tolerance and the minimum and maximum acceptable values.**

Colors	Nominal	Tolerance	Minimum	Maximum
red-red-black silver	22@10% Ω	10%	19.5 Ω	24.5 Ω
blue-gray black-gold	68@5% Ω	5%	65.4 Ω	71.4 Ω
brown-green brown-gold	150@5% Ω	5%	142.5 Ω	157.5 Ω
orange-orange brown-silver	330@10% Ω	10%	297 Ω	363 Ω
green-blue brown – gold	560@5% Ω	5%	533 Ω	588 Ω
brown-red red–silver	1200@10% Ω	10%	1180Ω	1320 Ω
red-violet-red– silver	2700@10% Ω	10%	2430 Ω	2970 Ω
gray-red-red– gold	8200@5% Ω	5%	7780 Ω	8690 Ω
brown-black orange–gold	10000@5% Ω	5%	95000 Ω	10500 Ω
orange-orange orange–silver	33000@10%	10%	29700 Ω	36300 Ω
blue-gray yellow–none	680000@20%Ω	20%	544000 Ω	816000 Ω
green-black green-silver	5000000@10%Ω	10%	4500000 Ω	5500000 Ω

3. **Obtain five assorted resistor values and list them in the table below. Determine the minimum and maximum acceptable values based on the nominal value and tolerance. Record these values in the given Table. Using the DMM measure the actual value of the resistor and record it. Determine the deviation percentage of this component. (The deviation percentage may be found via: Deviation = 100 * (measured-nominal)/nominal.** ***Circle the deviation if the resistor is out of tolerance.*****)**

		
3900 Ω	**2700 Ω**	**2000 Ω**

	
3300 Ω	**100 Ω**

[These pictures are same for Task3 and task 4(Only gold(5%))]

Value	Value	Maximum	Minimum	Measured	Deviation
Orange-white-red-gold	3900 Ω	4095 Ω	3705 Ω	3860Ω	40 Ω
Red-purple-red-gold	2700 Ω	2835 Ω	2565 Ω	2680 Ω	20 Ω
Red-black-red-gold	2000 Ω	2100 Ω	1900 Ω	2045 Ω	55Ω
Orange-orange-red-gold	3300 Ω	3465 Ω	3135 Ω	3248 Ω	52 Ω
Brown-black-brown-gold	100 Ω	105 Ω	95 Ω	98 Ω	2 Ω

4. Select ten resistors (5% tolerance) between 1Ω and 1MΩ. Verify the value of resistance and tolerance of resistors using color codes, and complete the following table.

S. No.	Resistor's Color Code *(Record four color bands)*	Coded Resistance (Ω)	Tolerance (%)	Maximum Coded Resistance (Ω)	Minimum Coded Resistance (Ω)	Measured Resistance (Ω)	Is the Resistor Within Tolerane ?
1	**Orange -brown Orange-gold**	31kΩ	5%	31k+3.1 .k = 34.1kΩ	31k-3.1k =27.9kΩ	28.1 kΩ	YES**
2	**Orange-white-red-gold**	3900 Ω	5%	3900+195= 4095 Ω	3900-195=37 05 Ω	3.86kΩ	YES**
3	**Red-purple-red-gold**	2700 Ω	5%	135+ 2700= 2835 Ω	135-2700= 2565 Ω	2.68k Ω	YES**
4	**Red-black-red-gold**	2000 Ω	5%	100+ 2000= 2100 Ω	100-2000= 1900 Ω	2.045k Ω	YES**
5	**Orange-orange-red-gold**	3300 Ω	5%	165+ 3300 3465 Ω	165-3300 3135Ω	3.248k Ω	YES**
6	**Brown-black-brown-gold**	100 Ω	5%	100+5= 105 Ω	100+5= 95 Ω	0.980k Ω	YES**
7	**blue-gray yellow–none**	680000 **Ω**	20%	816000 **Ω**	544000 **Ω**	574k **Ω**	YES**
8	**Green-black Green-silver**	500000 **Ω**	10%	550000 0 **Ω**	450000 0 **Ω**	510k **Ω**	YES**

Conclusion:

1. Resistors are essential electrical components that restrict the flow of current, with resistance measured in Ohms.
2. The actual resistance of a resistor may vary from its nominal value due to manufacturing tolerances.
3. Precision resistors (1% tolerance or better) have their values printed, while general-purpose resistors use color codes.
4. The resistor color code system, standardized by the EIA, assigns numerical values to colors for easy identification.
5. The first two or three color bands represent significant digits, while the next band indicates the multiplier.
6. The tolerance band specifies the allowable variation in resistance, typically given as a percentage.
7. Understanding resistor color coding is crucial for accurate circuit design and component selection.